

Homework 3

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Exercise 1

Determine the homeomorphism h showing that the following systems are topologically conjugate.

$$\begin{cases} \dot{x}_1 = -x_1 \\ \dot{x}_2 = x_2 \end{cases} \quad \begin{cases} \dot{y}_1 = -y_1 + y_2 \\ \dot{y}_2 = 2y_2 \end{cases}$$

Solution: As hinted we will first provide a homeomorphism g that will take the first system to the system

$$\begin{cases} \dot{z}_1 = -z_1 \\ \dot{z}_2 = 2z_2 \end{cases}.$$

We have seen this case before, so without going into too much detail, it can be verified that $g(p) = (g_1(p), g_2(p))$ defined as $g_1(p_1, p_2) = p_1$, $g_2(p_1, p_2) = \frac{p_2^3}{|p|} = p \cdot |p| = p^2 \operatorname{sgn}(p)$ is a homeomorphism that maps the x -system into the z -system.

Now we just need to provide a homeomorphism that maps the z -system into the y -system. First, notice that both these systems are linear with the matrices $D = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 1 \\ 0 & 2 \end{pmatrix}$, and $\dot{z} = Dz$, $\dot{y} = By$.

As continuous linear dynamical systems, we know that they are C^k -conjugate ($k \geq 1$) if and only if their defining matrices are similar.

After some calculations, we can find that the matrix $A = \begin{pmatrix} 1 & \frac{1}{3} \\ 0 & 1 \end{pmatrix}$ is such that $BA = AD$, that is, B and D are similar.

Since the z -system and the y -system are C^k -conjugate, they are also C^0 -conjugate, thus $h(p) = A(g(p))$ is a homeomorphism between the original two systems. $\square$

Exercise 2

Show that C^0 -equivalence divides the set $L(\mathbb{R}^4)$ into infinitely many classes.

Solution: By Kuiper's theorem two matrices $A, B \in L(\mathbb{R}^4)$, where $c(A) = c(B) = 4$ are C^0 -equivalent if and only if they are linearly equivalent.

We will show a construction where we have a family of infinitely many linearly non equivalent matrices that all have $c(A_n) = 4$.

Let us take the last example of the $L(\mathbb{R}^2)$ C^0 -equivalence classes $B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. From this 2×2 matrix we can construct a 4×4 matrix like $A_n = \begin{pmatrix} B & 0 \\ 0 & nB \end{pmatrix}$.

The eigenvalues of B are $\pm i$, whereas the eigenvalues of nB are $\pm ni$. Since A_n is a block matrix with blocks B and nB , its eigenvalues are therefore $\{-i, +i, -ni, +ni\}$. We can clearly see that all of the eigenvalues for any $n > 0$ have zero real part, thus in fact $c(A_n) = 4 \quad \forall n > 0$.

We have shown that to any A_n the Kuiper theorem applies, now we just need to show that A_n and A_m are linearly independent for any $n \neq m$.

A_n and A_m are linearly equivalent if $\exists \alpha, P$ such that $A_n = \alpha P A_m P^{-1}$. Given that A_n and A_m have the same upper left block B , we can conclude that α must be 1. Now we just need to show that A_n and A_m are not similar.

We know that if two matrices are similar then they share the same eigenvalues, however, in our case $\lambda(A_n) = \{\pm i, \pm ni\}$ and $\lambda(A_m) = \{\pm i, \pm mi\}$, that is $\lambda(A_n) \neq \lambda(A_m)$, therefore the two cannot be similar.

We have just shown a construction where $c(A_n) = 4$ and A_n and A_m are not similar for any $n \neq m$, thus Kuiper's theorem tells us that all A_n are in different classes for all $n > 0$, and since there are (countably) infinitely many choices for n there are (countably) infinitely many C^0 -equivalence classes in $L(\mathbb{R}^4)$.

Note, that this argument can be extended to use $x \in \mathbb{R}^+$ instead of $n \in \mathbb{Z}^+$ to give us uncountably infinitely many equivalence classes. □

Exercise 3

In which classes (according to C^0 -equivalence) is the following system as λ runs in $\mathbb{R}$?

$$\begin{cases} \dot{x}_1 = x_1 + \lambda x_2 \\ \dot{x}_2 = \lambda x_1 + \lambda x_2 \end{cases}$$

Solution: It is clear to see that we can write this system as $\dot{x} = Ax$ where $A = \begin{pmatrix} 1 & \lambda \\ \lambda & \lambda \end{pmatrix}$. Using Kuiper's theorem we just need to analyze the eigenvalue structure of this simple matrix to get a complete view of the C^0 -equivalence classes.

To calculate the eigenvalues we need to find the roots of the matrix's characteristic polynomial:

$$P_A(x) = (x-1)(x-\lambda) - \lambda^2 = x^2 - (1+\lambda)x + \lambda - \lambda^2 = 0$$
$$x_{1,2} = \frac{1+\lambda \pm \sqrt{(1+\lambda)^2 - 4\lambda + 4\lambda^2}}{2} = \frac{1+\lambda \pm \sqrt{5\lambda^2 - 2\lambda + 1}}{2}$$

We can clearly see that when $\lambda \in \mathbb{R}$ the determinant of the above expression is always positive, so we do not have to worry about complex eigenvalues, meaning that $\text{Re}(x) = x$.

According to Kuiper's theorem we need only know the sign of the eigenvalues to characterize the C^0 -equivalence. To find the sign we can solve the equation $x_{1,2}(\lambda) = 0$.

After some calculations we can come to the conclusion that the original system has a zero eigenvalue only in the cases where $\lambda = 0$ or $\lambda = 1$. This splits the real line into 5 components that need to be treated: $\lambda < 0$, $\lambda = 0$, $\lambda \in (0, 1)$, $\lambda = 1$, $\lambda > 1$.

(From now we will use the convention that x_1 is the root of a quadratic which we get by using the plus in the $\pm$ of the formula.)

1. $\lambda < 0$: $x_1 < 0$, $x_2 > 0$.
2. $\lambda > 1$: $x_1 > 0$, $x_2 < 0$.
3. $\lambda = 0$: $x_1 = 0$, $x_2 = 1$.
4. $\lambda = 1$: $x_1 = 0$, $x_2 = 2$.
5. $\lambda \in (0, 1)$: $x_1 > 0$, $x_2 > 0$.

From this we can conclude that cases 1. and 2. are equivalent, and cases 3. and 4. are also equivalent.

Thus we get three classes:

1. $\lambda \in (-\infty, 0) \cup (1, +\infty)$: $A \sim \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ (saddle)
2. $\lambda \in \{0, 1\}$: $A \sim \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ (infinitely many equilibrium points)
3. $\lambda \in (0, 1)$: $A \sim \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (source)

□